

COURSE TITLE : TOOL ENGINEERING LAB – I
COURSE CODE : 2119
COURSE CATEGORY : P
PERIODS/WEEK : 3
PERIODS/SEMESTER : 45
CREDITS : 2

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Practice on Tool room Lathe	9
2	Practice on Shaper	9
3	Practice on Drilling Machine	9
4	Practice on Milling Machine	9
5	Bench Work	9
	Total	45

COURSE OUTCOMES

On completion of the course, Student will be able to:

1. Perform the various operations on machine tools.
2. Perform special turning operation to produce male and female fit assembly, nut & bolts.
3. Know various operations on shaper to produce flat, step and V-shapes.
4. Perform the drilling, reaming, counter boring and countersinking operations on various drilling machine.
5. Understand various operations on milling machine to produce flat, step and channels.
6. Calculate the tap hole size, nominal diameter of bolt to perform tapping and dying operations on a job.

CONTENT OUTLINES

1. Practice on Tool Room Lathe-Cylindrical hole & shaft matching, Nut & bolt
2. Practice on Shaper -Shaping a rectangular plate, Step shaping, V shape
3. Practice on Drilling machine(Sensitive & Radial)- Counter boring, Counter sinking, Reaming,
4. Practice on Milling machine – Plain milling, Step milling, Channel milling
5. Bench work- Tapping , External Thread Cutting, Drilling, Reaming , Filing practice